

**B.Tech. Degree III Semester Regular/Supplementary Examination
November 2023**

CS/IT 19-202-0303/19-204-0303 DISCRETE COMPUTATIONAL STRUCTURES
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Use logical notation to define and reason mathematically about the fundamental data types and structures used in computer algorithms.
- CO2: Summarise mathematical notations and concepts in discrete mathematics that is essential for computing.
- CO3: Construct proofs using direct proof, proof by contraposition, proof by contradiction and proof by resolution and by mathematical induction.
- CO4: Familiarise mathematical reasoning and proof strategies.
- CO5: Identify and apply the counting principle.
- CO6: Apply graph theory to solve real world problems.
- CO7: Interpret the conceptual background needed to be able to identify structures of algebraic nature, and discover, prove and use properties about them.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer **ALL** questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a)	Show that $(p \rightarrow q)$ is logically equivalent to $(\neg p \vee q)$ using truth table.	3	L1	2	1,2
(b)	Translate the following statements in proposition logic:	3	L2	3	1,2
(i)	If you study, you will get good marks. If you do not study, you will enjoy. Therefore, either you will get good marks or you will enjoy.				
(ii)	If the catalogue is correct, then if the seeds are planted on April, flowers will bloom in July.				
(iii)	If John is elected class representative, then either Mary is elected treasurer or Alice is elected vice-treasurer.				
(c)	State Pigeon-hole Principle. How many people must you have to guarantee that at least 9 of them will have birthdays in the same day of the week?	3	L2	5	1,2
(d)	Select a theta notation for each expression and code segment.	3	L3	2	1
(i)	$(4n+1)^2$				
(ii)	$(5n+3)(1+\log n)$				
(iii)	for $i=1$ to $3n$ for $j=1$ to n $y=y+1$				
(e)	Draw a 3 regular graph with 4 vertices.	3	L2	6	1

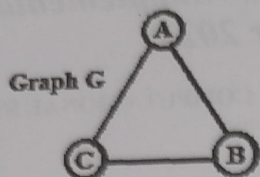
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Marks	BL	CO	PO
3	L2	6	1,2

- (f) Draw the spanning tree of the given graph.



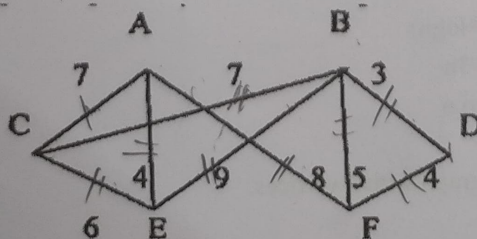
- (g) Define semigroup and monoids with examples.
(h) Explain the properties of a lattice.

3	L1	7	1
3	L1	7	1

PART B

(4 × 12 = 48)

- II. (a) Identify the domain and range of the relation.
 $R = \{(1,3), (2,2), (3,1), (4,0)\}$
- (b) If $A = \{x,y,z\}$, $B = \{1,2,3\}$, $C = \{x,y\}$, $D = \{2,3\}$ and the relations R and S are defined as $R: A \rightarrow B = \{(x,1), (x,2), (y,3)\}$, $S: C \rightarrow D = \{(x,2), (y,3)\}$, Find $R \cap S$, $R \cup S$, $R-S$, R'
- (c) Determine whether the relation $R = \{(a, b) \in R : a + b \text{ is even}\}$ on the set of positive integers, is an equivalence relation.
- OR
- III. (a) Determine the validity of the following argument without using truth tables. "Either I will pass the examination, or, I will not graduate. If I do not graduate, I will go to Australia. I failed: Thus I will go to Australia."
- (b) If $f, g, h : R \rightarrow R$ are defined by $f(x) = x^3 - 4x$, $g(x) = 1/(x^2 + 1)$ and $h(x) = x^4$, find $\{(f \circ g) \circ h\}(x)$ and $\{f \circ (g \circ h)\}(x)$ and check if they are equal.
- IV. (a) Evaluate Big-oh notation for the factorial function and the logarithm of the factorial function, where the factorial function $f(n) = n!$ is defined by $n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n$ where n is a positive integer.
- (b) Formulate recurrence relation which can define the number of subsets of an n element set.
- OR
- V. (a) Write the algorithm to perform bubble sort. Calculate its time complexity in terms of best case and worst case scenarios.
- (b) Solve recurrence relation $a_n - 7a_{n-1} + 10a_{n-2} = 0$ with initial condition $a_0 = 0$ and $a_1 = 6$.
- VI. (a) Define a bipartite and complete bipartite graph with example.
- (b) Find the minimum spanning tree using Kruskal's algorithm for the given graph

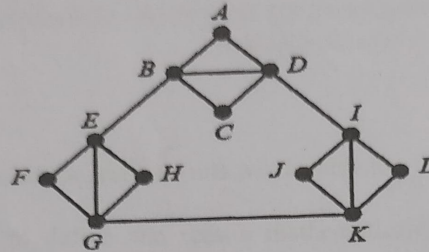


OR

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- | | | Marks | BL | CO | PO |
|----------|---|-------|----|----|------|
| VII. (a) | Give an example of a graph that has an Euler circuit which is also Hamiltonian. | 4 | L4 | 6 | 1,2 |
| (b) | Write the Fleury's algorithm and apply the same for the following graph. | 8 | L3 | 6 | 1,2, |



- | | | | | | |
|-----------|---|---|----|---|-----|
| VIII. (a) | Explain sublattice, isomorphic lattice and distributive lattice using suitable figures. | 6 | L2 | 7 | 1 |
| (b) | Draw the Hasse diagram representing the partial ordering $\{(a, b) a \text{ divides } b\}$ on $\{1, 2, 3, 4, 6, 8, 12\}$. | 6 | L2 | 7 | 1,2 |
| OR | | | | | |
| IX. (a) | Consider an algebraic system $(G, *)$ where G is the set of all non-zero real numbers and $*$ is a binary operation defined by $a*b = ab/2$. Show that G is abelian. | 5 | L3 | 7 | 1,2 |
| (b) | Let D be the set of divisors of 60 and divisibility $/$ be the partial ordering. Draw the Hasse diagram. Find the following | 7 | L4 | 7 | 1,2 |
| | (i) Upperbound, lowerbound, LUB and GLB of $S = \{12, 30\}$ | | | | |
| | (ii) Upperbound, lowerbound, LUB and GLB of $S = \{10, 15\}$ | | | | |

Bloom's Taxonomy Levels

L1 = 12.5%, L2 = 28.3%, L3 = 32.5%, L4 = 26.7%.
